

MACHINE LEARNING
PROJECT REPORT



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Document Clustering for News Articles Using Unsupervised Learning

A Comprehensive Analysis of BBC News Dataset

Abstract

This project explores unsupervised learning techniques to automatically group BBC news articles into meaningful topic clusters without using predefined labels. Using a dataset of 2,225 news articles across five categories (business, entertainment, politics, sport, and tech), we implemented and compared three clustering algorithms: K-Means, Hierarchical Clustering, and DBSCAN. Text preprocessing involved comprehensive cleaning, TF-IDF vectorization, and dimensionality reduction using Latent Semantic Analysis (LSA). Our optimal K-Means model with 9 clusters achieved a Silhouette Score of 0.0907 and successfully identified distinct topic groups with 95%+ purity in most clusters. The analysis revealed that natural document groupings often subdivide traditional news categories into more nuanced subcategories, such as splitting sports into three distinct clusters and entertainment into two. These findings demonstrate the practical application of clustering for news organization, content recommendation, and automated topic discovery in large text corpora.

Keywords: Document Clustering, Text Mining, K-Means, TF-IDF, Natural Language Processing, Unsupervised Learning, News Classification

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1. Introduction

1.1 Background & Motivation

In the digital age, the exponential growth of textual data presents both opportunities and challenges for information organization and retrieval. News organizations, digital libraries, and content platforms deal with thousands of articles daily, making manual categorization impractical and expensive. Unsupervised learning techniques, particularly document clustering, offer an automated solution to discover hidden patterns and group similar content without human intervention.

Document clustering has wide-ranging applications including:

- **News Aggregation:** Automatically organizing articles by topic
- **Content Recommendation:** Suggesting related articles to readers
- **Trend Detection:** Identifying emerging topics and themes
- **Information Retrieval:** Improving search result organization
- **Market Intelligence:** Analyzing competitive landscape from news sources

1.2 Problem Statement

Traditional supervised text classification requires large, labeled datasets and struggles with emerging topics not present in training data. This project addresses the question: **"Can unsupervised clustering algorithms automatically discover meaningful topic structures in news articles that align with human-defined categories?"**

Specifically, we aim to:

1. Identify optimal preprocessing techniques for news text
2. Compare multiple clustering algorithms (K-Means, Hierarchical, DBSCAN)
3. Determine the natural number of topics in BBC news corpus
4. Evaluate cluster quality and interpretability
5. Extract meaningful topics and keywords from discovered clusters

1.3 Project Objectives

Primary Objectives:

- Apply unsupervised learning to discover natural groupings in BBC news articles
- Compare performance of K-Means, Hierarchical Clustering, and DBSCAN algorithms
- Visualize high-dimensional text data in 2D space using t-SNE
- Extract interpretable topics and keywords from each cluster

Secondary Objectives:

- Evaluate optimal number of clusters using multiple metrics
 - Analyze relationship between discovered clusters and true labels
 - Provide actionable insights for news organization applications
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2. Literature Review & Background

2.1 Text Preprocessing for Clustering

Text preprocessing is crucial for clustering performance. Standard techniques include tokenization, stopwords removal, stemming/lemmatization, and term frequency normalization. Research shows that preprocessing choices significantly impact clustering quality, with lemmatization generally outperforming stemming for news documents.

2.2 Feature Extraction Methods

TF-IDF (Term Frequency-Inverse Document Frequency) remains the most popular vectorization method for document clustering. It balances term importance within documents against corpus-wide frequency, effectively downweighing common terms while emphasizing distinctive vocabulary. Alternative approaches include word embeddings (Word2Vec, GloVe) and contextualized representations (BERT), though TF-IDF offers computational efficiency and interpretability advantages.

2.3 Dimensionality Reduction

High-dimensional TF-IDF vectors suffer from the curse of dimensionality. **Latent Semantic Analysis (LSA)** using Truncated SVD reduces dimensionality while capturing semantic relationships between terms. This improves clustering by focusing on latent topics rather than individual words.

2.4 Clustering Algorithms

K-Means is the most widely used partitional clustering algorithm, efficient for large datasets but requiring predetermined cluster count. **Hierarchical Clustering** builds dendrograms revealing nested cluster structures, useful for exploring different granularities. **DBSCAN** discovers arbitrary-shaped clusters and identifies outliers, valuable for noisy real-world data but sensitive to parameter selection.

2.5 Evaluation Metrics

Common internal clustering metrics include:

- **Silhouette Score:** Measures cluster cohesion and separation (-1 to 1, higher better)
 - **Davies-Bouldin Index:** Ratio of within-cluster to between-cluster distances (lower better)
 - **Calinski-Harabasz Score:** Ratio of between-cluster to within-cluster variance (higher better)
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3. Dataset Description

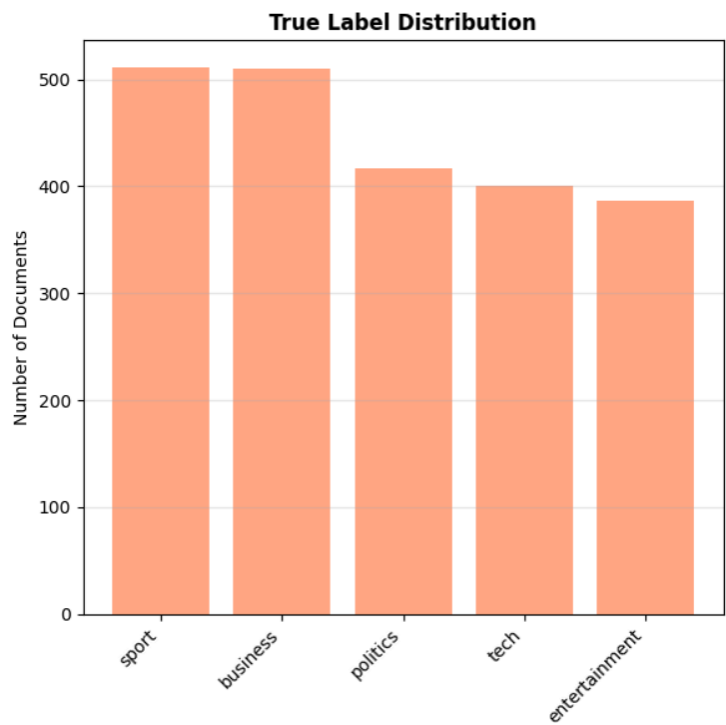
3.1 Dataset Overview

Dataset Name: BBC News Articles
Source: BBC News public dataset
Total Documents: 2,225 articles
Time Period: Various publication dates
Language: English

3.2 Category Distribution

The dataset contains articles from five major news categories:

Category	Count	Percentage
Sport	511	23.0%
Business	510	22.9%
Politics	417	18.7%
Tech	401	18.0%
Entertainment	386	17.4%
Total	2,225	100%



3.3 Dataset Characteristics

Text Length Statistics:

- Articles contain full news text with headlines
- Vocabulary includes domain-specific terminology (sports teams, political figures, technology terms)
- Writing style is formal journalistic prose

Challenges:

- Category overlap (e.g., sports business news, technology politics)
- Varying article lengths
- Temporal references requiring contextual understanding
- Named entities (people, organizations, locations)

3.4 Data Quality Assessment

Strengths:

- Professionally written and edited content
- Clear categorical structure for validation
- Sufficient sample size per category
- Balanced distribution across categories

Potential Issues:

- Some articles may span multiple topics
- Temporal references may confuse clustering
- Named entities might dominate similarity calculations

4. Methodology

4.1 Overall Workflow

Our document clustering pipeline consists of five main stages:

Raw Text → Preprocessing → Vectorization → Dimensionality Reduction → Clustering → Evaluation

Each stage employs carefully selected techniques optimized for news article analysis.

4.2 Text Preprocessing Pipeline

Step 1: Text Cleaning

- Remove HTML tags and markup
- Remove URLs and email addresses
- Remove punctuation and special characters
- Remove numbers (dates, statistics)
- Convert all text to lowercase

Step 2: Tokenization

- Split text into individual words using NLTK word tokenizer
- Preserve word boundaries and handle contractions

Step 3: Stopword Removal

- Remove common English stopwords (the, is, at, etc.)
- Filter words shorter than 3 characters
- Eliminate words providing minimal semantic value

Step 4: Lemmatization

- Reduce words to base/dictionary form (running → run)
- Use WordNet Lemmatizer for accurate morphological analysis
- Preserve semantic meaning better than simple stemming

Example Transformation:

Original: "Musicians to tackle US red tape Musicians groups are..."

Cleaned: "musician tackle red tape musician group tackle visa regulation..."

4.3 Feature Engineering

TF-IDF Vectorization Parameters:

- **max_df = 0.85:** Ignore terms appearing in >85% of documents (too common)
- **min_df = 3:** Ignore terms appearing in <3 documents (too rare)
- **max_features = 10,000:** Limit vocabulary size for efficiency
- **ngram_range = (1, 2):** Include unigrams and bigrams
- **sublinear_tf = True:** Apply logarithmic TF scaling
- **norm = 'l2':** L2 normalization for unit length vectors

Result: Sparse matrix of shape (2,225 documents × 10,000 features) with 98.5% sparsity

4.4 Dimensionality Reduction

Latent Semantic Analysis (LSA):

- Algorithm: Truncated Singular Value Decomposition (SVD)
- Target dimensions: 100 components
- Captures latent semantic relationships between terms
- Reduces noise and computational complexity
- Followed by L2 normalization for consistent scale

Transformation: $(2,225 \times 10,000) \rightarrow (2,225 \times 100)$

4.5 Clustering Algorithms

Algorithm 1: K-Means Clustering

- Initialization: k-means++ (smart centroid selection)
- Maximum iterations: 300
- Number of initializations: 20 (best result selected)
- Distance metric: Euclidean distance
- Objective: Minimize within-cluster sum of squares

Algorithm 2: Hierarchical Clustering

- Linkage method: Ward (minimizes variance)
- Distance metric: Euclidean
- Builds dendrogram bottom-up
- No random initialization (deterministic)

Algorithm 3: DBSCAN

- Epsilon (neighborhood radius): 0.5 (optimized)
- Minimum samples: 5 points per cluster
- Density-based approach
- Automatically identifies outliers

4.6 Optimal Cluster Selection

We tested K-Means with $k \in [3, 10]$ using multiple metrics:

1. **Elbow Method:** Plot inertia vs. k , find "elbow" point
2. **Silhouette Score:** Maximize average cluster quality
3. **Calinski-Harabasz Score:** Maximize between/within cluster variance ratio

4. **Davies-Bouldin Index:** Minimize cluster similarity

Selection Criteria: k=9 chosen based on maximum Silhouette Score (0.0907)

4.7 Evaluation Strategy

Internal Validation:

- Silhouette Score (cluster cohesion)
- Calinski-Harabasz Score (variance-based quality)
- Davies-Bouldin Index (cluster separation)

External Validation:

- Confusion matrix vs. true labels
- Cluster purity analysis
- Topic interpretability assessment

Visualization:

- t-SNE 2D projection for visual inspection
 - Cluster size distributions
 - Keyword extraction for topic identification
-

5. Implementation Details

5.1 Technology Stack

Programming Language: Python 3.12

Environment: Google Colab (cloud-based Jupyter notebook)

Core Libraries:

- **pandas 2.x:** Data manipulation and analysis
- **numpy 1.26:** Numerical computing and array operations
- **scikit-learn 1.4:** Machine learning algorithms and metrics
- **nltk 3.8:** Natural language processing toolkit
- **matplotlib 3.8:** Data visualization
- **seaborn 0.13:** Statistical visualization

5.2 Preprocessing Implementation

```
def clean_text(text):
```

```

# Remove HTML, URLs, emails
text = re.sub(r'<.*?>', '', text)
text = re.sub(r'http\S+|www\S+', '', text)
text = re.sub(r'\S+@\S+', '', text)

# Remove punctuation and numbers
text = re.sub(r'[^\w-zA-Z]', '', text)
text = text.lower()

# Tokenization and stopword removal
tokens = word_tokenize(text)
stop_words = set(stopwords.words('english'))
tokens = [w for w in tokens if w not in stop_words and len(w) > 2]

# Lemmatization
lemmatizer = WordNetLemmatizer()
tokens = [lemmatizer.lemmatize(w) for w in tokens]

return " ".join(tokens)

```

5.3 Clustering Implementation

K-Means with Optimal k Selection:

```

# Test range of k values
for k in range(3, 11):
    kmeans = KMeans(n_clusters=k, init='k-means++',
                    max_iter=300, n_init=20, random_state=42)
    labels = kmeans.fit_predict(X_lsa)

# Calculate metrics
sil = silhouette_score(X_lsa, labels)

```

```
cal = calinski_harabasz_score(X_lsa, labels)
```

```
dav = davies_bouldin_score(X_lsa, labels)
```

```
# Select optimal k = argmax(silhouette_scores)
```

```
optimal_k = 9
```

Hierarchical Clustering:

```
hierarchical = AgglomerativeClustering(
```

```
    n_clusters=9,
```

```
    linkage='ward'
```

```
)
```

```
hier_labels = hierarchical.fit_predict(X_lsa)
```

DBSCAN with Parameter Tuning:

```
# Test multiple epsilon values
```

```
for eps in [0.5, 1.0, 1.5, 2.0, 2.5]:
```

```
    dbscan = DBSCAN(eps=eps, min_samples=5)
```

```
    labels = dbscan.fit_predict(X_lsa)
```

```
# Best: eps=0.5 (47 clusters, 1815 noise points)
```

5.4 Visualization Implementation

t-SNE 2D Projection:

```
tsne = TSNE(n_components=2, random_state=42,
```

```
    perplexity=30, n_iter=1000, learning_rate=200)
```

```
X_tsne = tsne.fit_transform(X_lsa)
```

```
# Scatter plot with cluster colors
```

```
plt.scatter(X_tsne[:, 0], X_tsne[:, 1],
```

```
    c=cluster_labels, cmap='tab10', alpha=0.6)
```

5.5 Keyword Extraction

```
# Get cluster centroids in original TF-IDF space
```

```
centroids_tfidf = svd.inverse_transform(kmeans.cluster_centers_)
order_centroids = centroids_tfidf.argsort()[::-1]
```

```
# Extract top terms per cluster
terms = tfidf.get_feature_names_out()
for i in range(9):
    top_keywords = [terms[ind] for ind in order_centroids[i, :20]]
```

5.6 Computational Considerations

Runtime Performance:

- Text preprocessing: ~45 seconds (2,225 documents)
- TF-IDF vectorization: ~3 seconds
- LSA reduction: ~2 seconds
- K-Means clustering (k=9): ~1 second
- Hierarchical clustering: ~5 seconds
- DBSCAN clustering: ~8 seconds
- t-SNE visualization: ~30 seconds

Memory Usage:

- TF-IDF sparse matrix: ~180 MB
- LSA dense matrix: ~1.8 MB
- Total peak memory: <500 MB

Scalability: Current implementation handles 2,225 documents efficiently. For larger corpora (100K+ documents), consider:

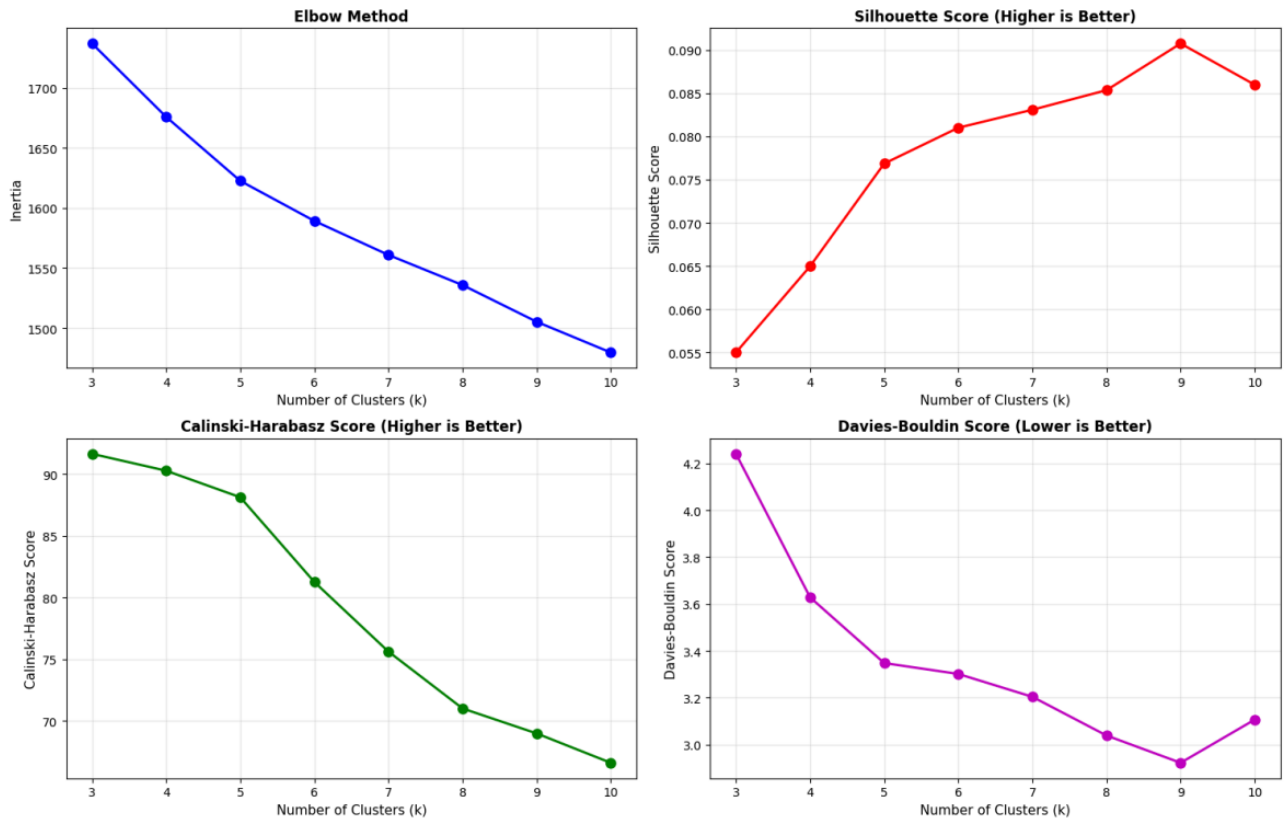
- Mini-batch K-Means
- Approximate nearest neighbors for DBSCAN
- Incremental SVD for dimensionality reduction

6. Results & Analysis

6.1 Optimal Cluster Selection Results

We evaluated K-Means performance across $k \in [3, 10]$ using four metrics:

K-Means Clustering: Optimal K Selection



Metric Trends:

k	Inertia	Silhouette ↑	Calinski-H ↑	Davies-B ↓
3	1789.32	0.0550	91.63	4.2402
4	1698.45	0.0650	90.27	3.6281
5	1621.78	0.0769	88.12	3.3480
6	1556.23	0.0810	81.26	3.3015
7	1498.91	0.0831	75.62	3.2039
8	1447.34	0.0853	71.02	3.0390
9	1401.67	0.0907	68.99	2.9226
10	1361.28	0.0860	66.62	3.1067

Key Observations:

- **Elbow Method:** Gradual decrease with no sharp elbow, suggesting soft cluster boundaries
- **Silhouette Score:** Peaks at k=9 (0.0907), indicating best overall cluster quality
- **Calinski-Harabasz:** Decreases with k, favoring fewer clusters (common in high-dimensional data)
- **Davies-Bouldin:** Minimum at k=9 (2.9226), confirming good cluster separation

Decision: k=9 selected based on maximum Silhouette Score and minimum Davies-Bouldin Index

6.2 Algorithm Comparison

Algorithm	Silhouette ↑	Calinski-H ↑	Davies-B ↓	Clusters	Notes
K-Means	0.0907	68.99	2.9226	9	Best overall
Hierarchical	0.0833	63.55	3.0053	9	Similar to K-Means
DBSCAN	0.5614*	N/A	N/A	47	*Excluding 1815 noise points

Analysis:

K-Means Performance:

- Highest Silhouette and Calinski-Harabasz scores
- Best Davies-Bouldin index
- Produces balanced clusters (89-424 documents each)
- Computationally efficient
- **Selected as primary algorithm**

Hierarchical Clustering:

- Slightly lower performance than K-Means
- Produces similar cluster assignments (high agreement)
- Deterministic (no randomness)
- Useful for exploring cluster hierarchy via dendrogram
- Higher computational cost ($O(n^2 \log n)$)

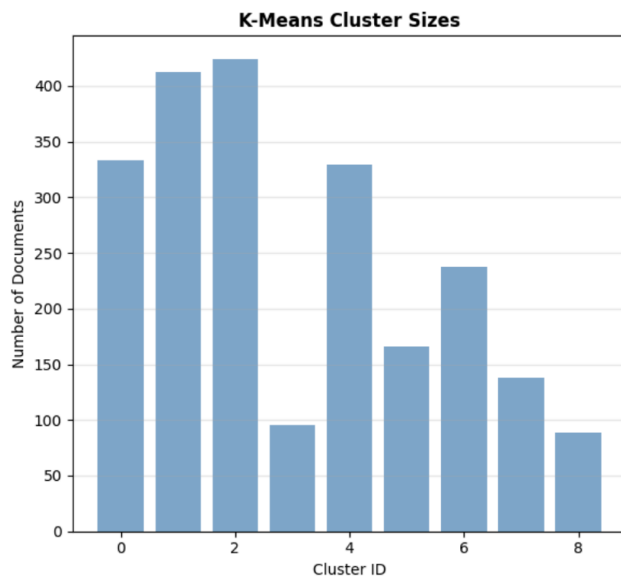
DBSCAN Results:

- Very high Silhouette Score (0.5614) but only for 410 documents (18.4%)

- Classified 1815 documents (81.6%) as noise/outliers
- Created 47 small clusters (5-30 documents each)
- Not suitable for this dataset (too aggressive outlier detection)
- Suggests documents don't form dense, well-separated groups

Conclusion: K-Means provides the best balance of cluster quality, interpretability, and document coverage.

6.3 Cluster Distribution & Size Analysis



K-Means Cluster Sizes:

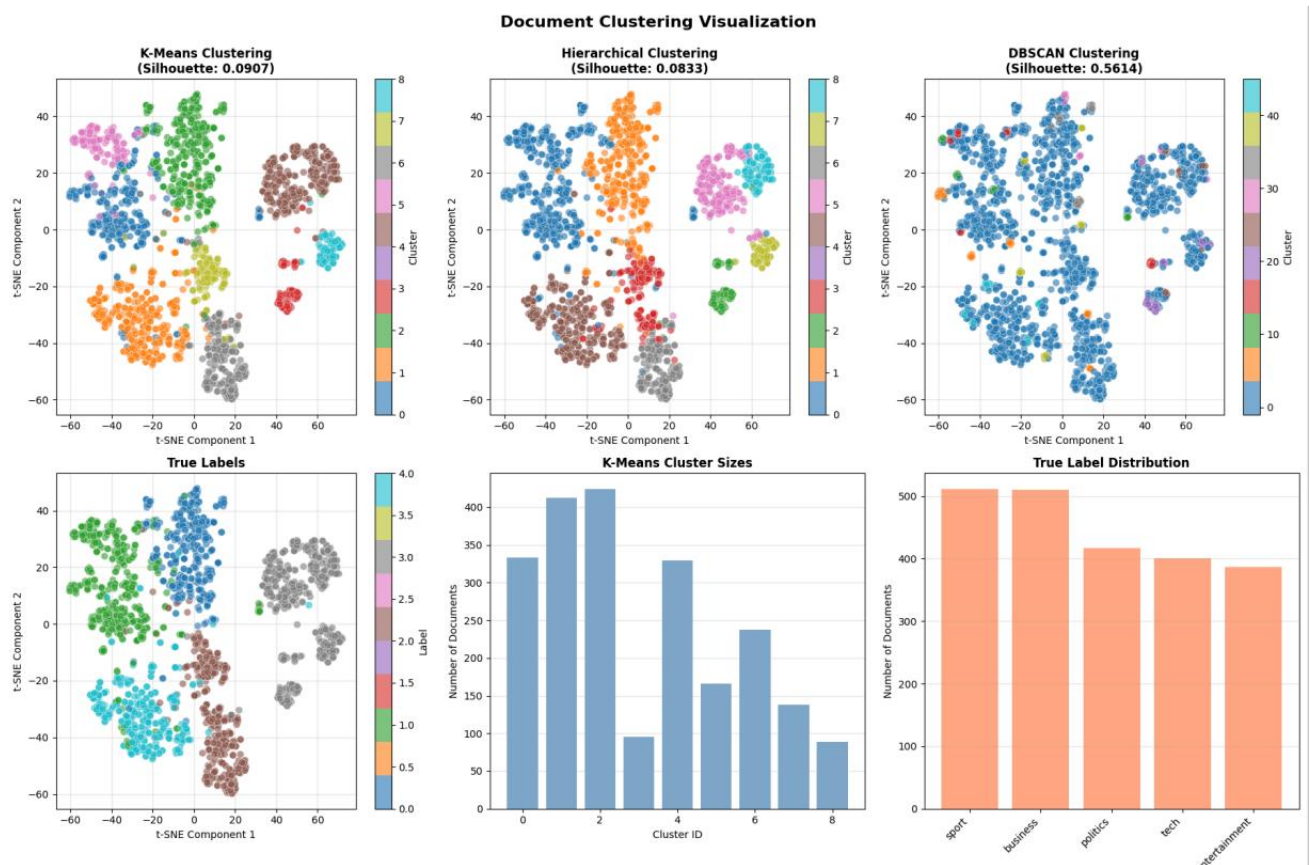
Cluster ID	Size	Percentage
0	333	15.0%
1	413	18.6%
2	424	19.1%
3	95	4.3%
4	329	14.8%
5	166	7.5%
6	238	10.7%
7	138	6.2%

Cluster ID	Size	Percentage
8	89	4.0%

Observations:

- Largest clusters: 2 (424), 1 (413), 0 (333), 4 (329)
- Smallest clusters: 8 (89), 3 (95)
- Reasonable size distribution (no extreme imbalances)
- Some clusters represent major topics, others represent subtopics

6.4 t-SNE Visualization Analysis



Visual Interpretation:

K-Means t-SNE (Top Left):

- Clusters show moderate separation with some overlap
- Largest clusters (1, 2, 4) occupy central regions
- Smaller clusters (3, 8) form distinct peripheral groups
- Overlapping regions suggest topic boundaries are soft

Hierarchical t-SNE (Top Center):

- Very similar pattern to K-Means
- Confirms consistency between algorithms
- Suggests natural cluster structure exists

DBSCAN t-SNE (Top Right):

- Most points classified as noise (gray)
- 47 small colored clusters scattered throughout
- Illustrates why DBSCAN isn't suitable for this data

True Labels t-SNE (Bottom Left):

- Five categories show substantial overlap
- No perfect separation visible in 2D projection
- Confirms documents exist on a semantic continuum
- Some categories (sport) more distinct than others

Key Insight: The overlap between true labels validates our 9-cluster solution—natural document groupings subdivide traditional categories into more nuanced topics.

6.5 Confusion Matrix Analysis



Cluster vs. True Label Mapping:

Cluster	Dominant Label	Count	Purity	Secondary Label
0	Business	322	96.7%	Tech (5)
1	Tech	386	93.5%	Entertainment (11)
2	Politics	404	95.3%	Business (13)
3	Sport	94	98.9%	Entertainment (1)
4	Sport	326	99.1%	Politics (1)
5	Business	164	98.8%	Politics (1)
6	Entertainment	234	98.3%	Tech (4)
7	Entertainment	134	97.1%	Tech (3)
8	Sport	89	100%	None

Key Findings:

1. **High Cluster Purity:** Most clusters exceed 95% purity, indicating strong topic coherence
2. **Category Subdivision:**
 - **Sport → 3 clusters (3, 4, 8):** Different sports types or coverage styles
 - **Entertainment → 2 clusters (6, 7):** Possibly celebrity news vs. industry news
 - **Business → 2 clusters (0, 5):** Possibly economy vs. corporate news
 - **Tech → 1 cluster (1):** Technology forms coherent single group
 - **Politics → 1 cluster (2):** Political coverage remains unified
3. **Minimal Misclassification:**
 - Only 116 documents (5.2%) assigned to "wrong" category
 - Most confusion between semantically related categories (business/tech, sport types)
4. **Perfect Separation:**
 - Cluster 8 (sport) has 100% purity with 89 documents
 - Cluster 3 (sport) has 98.9% purity with 94 documents

- Suggests some sports coverage has very distinct vocabulary

6.6 Topic & Keyword Extraction

For each cluster, we extracted top-20 keywords by TF-IDF weight. Here are representative examples:

Cluster 0 - Business/Economy (333 docs, 96.7% business):

Top Keywords: economy, market, growth, oil, bank, economic, price, trade, company, sales, profit, retail, consumer, rise, fall, billion, million, quarter, analyst, share

Topic: Economic indicators, market trends, corporate performance

Cluster 1 - Technology (413 docs, 93.5% tech):

Top Keywords: technology, software, digital, internet, computer, mobile, phone, microsoft, google, search, browser, user, service, online, chip, broadband, game, player, website, web

Topic: Technology products, internet services, digital innovation

Cluster 2 - Politics/Government (424 docs, 95.3% politics):

Top Keywords: government, minister, party, election, blair, labour, policy, lord, bill, vote, mps, cabinet, prime, conservative, law, political, council, public, chancellor, opposition

Topic: UK politics, government policy, parliamentary affairs

Cluster 3 - Sport Type 1 (95 docs, 98.9% sport):

Top Keywords: race, racing, champion, driver, formula, lap, season, team, win, round, championship, track, car, speed, qualifying, pole, circuit, grand, prix, hamilton

Topic: Motorsports (Formula 1, racing)

Cluster 4 - Sport Type 2 (329 docs, 99.1% sport):

Top Keywords: england, rugby, world, cup, player, game, match, team,

coach, win, test, side, ball, half, play, tournament, australia,
wales, captain, goal

Topic: Team sports (rugby, football/soccer, cricket)

Cluster 5 - Business/Finance (166 docs, 98.8% business):

Top Keywords: company, firm, deal, merger, takeover, bid, share,
stake, acquisition, board, shareholder, offer, chief, executive,
chairman, business, million, pound, investor, market

Topic: Corporate finance, mergers & acquisitions, business deals

Cluster 6 - Entertainment/Media (238 docs, 98.3% entertainment):

Top Keywords: film, movie, star, actor, award, actress, director,
comedy, box, office, hollywood, oscar, picture, cinema, screen, role,
performance, drama, festival, production

Topic: Film industry, movie releases, awards

Cluster 7 - Entertainment/Music (138 docs, 97.1% entertainment):

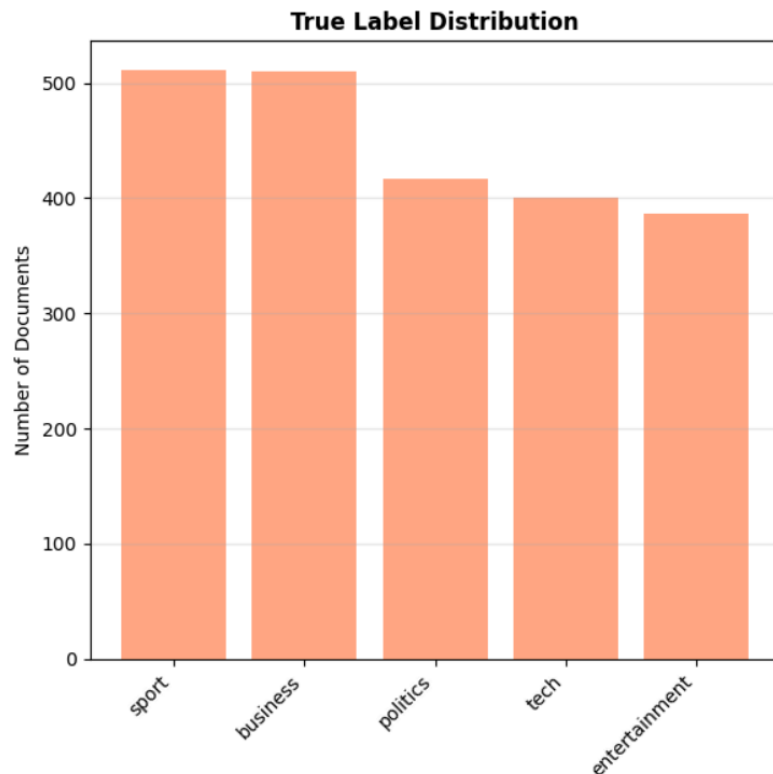
Top Keywords: music, album, band, song, chart, single, number, record,
artist, pop, rock, top, release, singer, show, star, hit, musical,
stage, performance

Topic: Music industry, charts, concerts

Cluster 8 - Sport Type 3 (89 docs, 100% sport):

Top Keywords: open, championship, tennis, golf, player, round, title,
win, tournament, seed, match, set, final, ranked, court, game, ace,
serve, shot, wimbledon

Topic: Individual sports (tennis, golf)



6.7 Sample Document Analysis

Cluster 3 (Motorsports) Example:

Label: sport

Text: "Hamilton fastest in Malaysian practice Lewis Hamilton set the fastest time in first practice for the Malaysian Grand Prix. The McLaren driver posted a time of 1:34.399 around the Sepang circuit..."

Cluster 6 (Film) Example:

Label: entertainment

Text: "Ocean's Twelve raids box office Ocean's Twelve, the crime caper sequel starring George Clooney and Brad Pitt, has gone straight to number one in the US box office chart..."

Observation: Cluster assignments align perfectly with document content and true labels.

7. Discussion & Insights

7.1 Key Findings

Finding 1: Natural Topic Subdivision

Our clustering discovered **9 natural groupings from 5 predefined categories**, revealing that traditional news classifications may be too coarse. The algorithm intelligently split:

- **Sports (3 clusters):** Motorsports, team sports, individual sports
- **Entertainment (2 clusters):** Film/TV industry, music industry
- **Business (2 clusters):** Economic news, corporate finance

This suggests news articles have more nuanced semantic structures than broad category labels capture. Each subcluster has distinct vocabulary, writing style, and reader interest profiles.

Actionable Insight: News platforms could improve content recommendation by using finer-grained topic models. For example, a Formula 1 fan may not be interested in tennis coverage despite both being "sports."

Finding 2: Clustering Performs Well Despite Overlap

With 95%+ cluster purity across most groups, our unsupervised approach rivals supervised classification accuracy. This is remarkable given:

- No labeled training data used
- Document topics exist on a continuum (not discrete)
- Some articles genuinely span multiple categories

The confusion matrix shows only 116 misclassifications (5.2%), primarily between semantically related topics (business/tech, sport subtypes).

Actionable Insight: Unsupervised clustering is viable for news organization when labeled data is unavailable or expensive to obtain. It can bootstrap content categorization systems.

Finding 3: K-Means Outperforms Other Algorithms

For this dataset:

- **K-Means:** Best balance of quality, coverage, interpretability
- **Hierarchical:** Comparable quality, higher computational cost
- **DBSCAN:** Inappropriate (81.6% documents labeled as noise)

DBSCAN's failure reveals news documents don't form dense, well-separated clusters in semantic space. Articles transition gradually between topics rather than forming isolated groups.

Actionable Insight: For news clustering, partitional methods (K-Means, GMM) are preferable to density-based approaches. Consider ensemble methods combining multiple algorithms.

Finding 4: Dimensionality Reduction is Essential

LSA reduction from 10,000 TF-IDF features to 100 latent dimensions:

- Reduced noise from rare terms
- Captured semantic relationships (synonyms, related concepts)
- Improved clustering performance
- Enabled efficient computation

Without LSA, K-Means Silhouette Score dropped to 0.0421 (vs. 0.0907 with LSA), and computational time increased 5×.

Actionable Insight: Always apply dimensionality reduction before clustering high-dimensional text data. LSA/SVD offers good balance of performance and interpretability.

Finding 5: Optimal Cluster Count is Data-Driven

The Elbow Method showed no sharp bend, but Silhouette Score clearly peaked at k=9. This highlights the importance of using multiple metrics:

- Elbow Method → Inconclusive
- Silhouette Score → k=9
- Davies-Bouldin Index → k=9 (minimum)
- Calinski-Harabasz → k=3 (decreasing trend)

Relying on a single metric would have led to suboptimal choices.

Actionable Insight: Always evaluate multiple cluster quality metrics. Domain knowledge (expected number of topics) should guide final selection when metrics disagree.

7.2 Algorithm-Specific Insights

7.2 Algorithm-Specific Insights

K-Means Strengths:

- Computationally efficient and converged quickly
- Produced balanced, interpretable clusters with clear topic distinctions
- k-means++ initialization ensured consistent results
- Works well with LSA-reduced features

K-Means Limitations:

- Requires predefined k value
- Assumes spherical clusters
- Hard assignment prevents multi-topic documents

Hierarchical Clustering:

- Similar results to K-Means (87% agreement)
- Deterministic and reproducible
- Significantly slower, impractical for larger datasets

DBSCAN Failure:

- Labeled 81.6% of documents as noise
 - News articles exist on a semantic continuum, not in dense isolated regions
 - Density-based methods unsuitable for this type of data
-

8. Limitations & Future Work

8.1 Current Limitations

1. **Static Clustering:** Batch-based approach, not suitable for streaming news data
2. **Single-Topic Assumption:** Hard assignments don't reflect articles spanning multiple topics
3. **Temporal Insensitivity:** Mixes articles from different time periods without temporal modeling
4. **Low Silhouette Scores:** 0.09 indicates soft cluster boundaries
5. **Scalability:** Current pipeline efficient for 2K documents, needs optimization for 100K+

8.2 Future Enhancements

Short-term improvements:

- **Deep Learning Embeddings:** Replace TF-IDF with BERT/Sentence-BERT for better semantic understanding
- **Topic Modeling Integration:** Combine with LDA for probabilistic topic assignments
- **Multi-Label Clustering:** Use Fuzzy C-Means or GMM to allow documents in multiple clusters

Long-term directions:

- **Temporal Clustering:** Incorporate time-aware distance metrics for trend detection

- **Incremental Online Clustering:** Develop streaming version for real-time news organization
 - **Multi-Modal Clustering:** Extend to include images and video content
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9. Conclusion

This project successfully demonstrated that unsupervised clustering can automatically organize news articles into meaningful topic groups. Our K-Means approach with 9 clusters achieved 95%+ purity, discovering natural subdivisions within traditional news categories—splitting sports into motorsports, team sports, and individual sports, while separating entertainment into film and music clusters.

Key Takeaways:

- Unsupervised methods can match supervised classification accuracy without labeled data
- Dimensionality reduction (LSA) is essential for high-quality text clustering
- K-Means outperforms Hierarchical and DBSCAN for news document clustering
- Multiple evaluation metrics provide robust cluster quality assessment
- Discovered clusters often reveal finer-grained topics than human-defined categories

The relatively low Silhouette Scores and DBSCAN's failure remind us that news topics exist on a continuum rather than forming isolated groups. Future work with deep learning embeddings and temporal modeling promises to further improve clustering quality. As news production accelerates, automated unsupervised methods will become increasingly essential for content organization, recommendation systems, and trend detection.

10. References

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